

Mathematical proof - Set 2

TMUA practice questions by topic. Specification Prf1-Prf4. 9 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 The sums 1 , $1 + 3$, $1 + 3 + 5$, $1 + 3 + 5 + 7$ are 1 , 4 , 9 , 16 . A student conjectures that the sum of the first n odd numbers is n^2 . Which of the following is a valid justification?
- A** Each odd number is one more than an even number, and the even numbers sum to $n(n - 1)$... so the total is n^2 since there are n of them: $n(n - 1) + n = n^2$.
- B** The conjecture is true because n^2 is always odd when n is odd.
- C** The pattern 1 , 4 , 9 , 16 continues with 25 , so the conjecture is true.
- D** The odd numbers form an arithmetic series with first term 1 and difference 2 , so the sum of n terms is $\frac{n}{2}(2 + 2(n - 1)) = n^2$.
- E** Four cases have been checked, which is enough for a pattern.
- 2 Which of the following statements can be disproved by a single counterexample?
- I** Every prime number greater than 2 is odd.
- II** Every integer of the form $4k + 3$ is prime.
- III** There exists an even prime number.
- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

- 3 Here is a proof that there are infinitely many primes, with the steps out of order.
- I Then N leaves remainder 1 on division by each of $p_1, \dots, p_n$, so no p_i divides N .
 - II Suppose there are only finitely many primes $p_1, p_2, \dots, p_n$, and let $N = p_1 p_2 \cdots p_n + 1$.
 - III So N has a prime factor that is not in the list, which is a contradiction.
 - IV But $N > 1$, so N has at least one prime factor.
- Which is the correct order?
- A II, I, IV, III
 - B II, III, I, IV
 - C I, II, IV, III
 - D II, I, III, IV
 - E IV, II, I, III
- 4 Consider the claim: **if x and y are real numbers with $x + y > 10$, then $x > 5$ or $y > 5$.** Which of the following proves it?
- A Suppose $x \leq 5$ and $y \leq 5$. Then $x + y \leq 10$, contradicting $x + y > 10$.
 - B Suppose $x > 5$. Then $y > 10 - x$, so $y > 5$.
 - C If $x + y > 10$ then the average of x and y is more than 5, so both exceed 5.
 - D Take $x = 6, y = 6$: then $x + y = 12 > 10$ and $x > 5$.
 - E If $x > 5$ then the conclusion holds; if $y > 5$ then the conclusion holds.
- 5 A student notices that $1 + 2 = 3, 4 + 5 + 6 = 7 + 8, 9 + 10 + 11 + 12 = 13 + 14 + 15$ and conjectures that the pattern continues. Which of the following describes the n th line of the pattern?
- A The n th line starts at n^2 and has $n + 1$ numbers on the left and n on the right.
 - B The n th line starts at $n(n + 1)$ and has $n + 1$ numbers on the left.
 - C The n th line starts at n^2 and has n numbers on the left and $n + 1$ on the right.
 - D The n th line starts at n^2 and has n numbers on each side.
 - E The n th line starts at $2n - 1$ and has $n + 1$ numbers on the left.

- 6** For a positive integer n it is known that: (i) if n is a multiple of 4 then n is not a multiple of 6; (ii) n is a multiple of 12 or n is a multiple of 4. What can be deduced?
- A** The premises are inconsistent unless n is a multiple of 4 but not of 6; in particular n cannot be a multiple of 12.
- B** n is not a multiple of 4.
- C** n is a multiple of 12.
- D** n is a multiple of 6.
- E** Nothing can be deduced.
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- 7** Let $u_n = n^2 - n + 41$. Which of the following statements is/are true?
- I** u_n is prime for $n = 1, 2, 3, 4, 5$.
- II** u_n is prime for all positive integers n .
- III** u_{41} is divisible by 41.
- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III
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- 8** Every element of a set S of integers is either a multiple of 3 or a multiple of 5. Every multiple of 3 in S is negative. S contains the number 10. Which of the following statements must be true?
- I** S contains a positive multiple of 5.
- II** S contains no positive multiple of 3.
- III** S contains a negative number.
- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

- 9** A student proves that $\sqrt{4}$ is irrational by copying the standard proof for $\sqrt{2}$: 'Suppose $\sqrt{4} = p/q$ in lowest terms; then $p^2 = 4q^2$, so p^2 is even, so p is even, $p = 2k$; then $4k^2 = 4q^2$, so $k^2 = q^2 \dots$ ' At which point does the argument break down compared with the $\sqrt{2}$ proof?
- A** $\sqrt{4}$ cannot be written as p/q .
- B** $p^2 = 4q^2$ does not imply that p is even.
- C** $k^2 = q^2$ does not show that q is even, so no contradiction arises.
- D** It is not valid to assume p and q have no common factor.
- E** The argument is valid, so $\sqrt{4}$ is irrational.